

Climate 5000 Series - Gen 4

12K BTU Single Zone Regular System

4-Way Cassette IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

12K Btu Single Zone Regular System - 4-Way Cassette IDU

Indoor part number	8-733-962-717
Indoor model number	BMS500-AAU012-1AHCXD
Outdoor part number	8-733-962-685
Outdoor model number	BMS500-AAS012-1CSXRD

Efficiency Data		AHRI 210/240 2023
SEER2		21.1
EER2		12.5
HSPF2 Region IV		10.4
HSPF2 Region V		7.9
COP(47F)		3.55
COP (5F)		2.40

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	3700~14000
	Rated Capacity	12000
	Current	4.3
	Input Power	968
Heating	Capacity Range	4700~15000
	Rated Capacity (47F)	12000
	Rated Capacity (5F)	8400
	Current	4.3
	Input Power	991

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 12.0
Max. fuse indoor/outdoor (A)	15.0 / 15.0

Compressor Data	
Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 310
Input (W)	325 / 826
Rated current (RLA/A)	2.40 / 5.65

Piping	
Refrigerant precharge (oz/kg)	32.45 / 0.92
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	1/4" / Ø6.35
Gas side (in/mm)	3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.16 / 15
Max. refrigerant pipe length (ft/m)	82.00 / 25
Max. difference in height (ft/m)	49.20 / 15

Operation Range

ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data		
Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		2.32
IDU	Fan motor model	ZKFN-45-8-2
	Fan motor qty (ea)	1
	Fan motor input (W)	50.0
	Fan speed (RPM)	700 / 600 / 400
	Air flow, Turbo / H / M / L / Si (CFM)	417.9 / 364.9 / 306.1 / 194.2 / 147.2
	Approx. Air Throw Range ft (m)	9.8 / 3
	Noise level, Turbo / H / M / L / Si (dB(A))	46 / 42.0 / 38.5 / 26.5 / 24.0
	Net weight (lbs / kg)	35.27 / 16.0
	Gross weight (lbs / kg)	41.89 / 19.0
	Panel net weight (lbs / kg)	5.95 / 2.7
ODU	Gross panel weight (lbs / kg)	9.48 / 4.3
	Drain pipe diameter (mm)	ODΦ25mm
	Fan motor model	ZKFN-34-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed (RPM)	800 / 700 / 600
	Air flow, Max (CFM)	1236.1
	Noise level, Max (dB(A))	54.5
	Net weight (lbs / kg)	62.17 / 28.2
	Gross weight (lbs / kg)	67.68 / 30.7

Accessories	
BMS-CCG-XXD (8733962708)	Decorative Grille - 9K ~ 18K
BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK2-7X1 (8733951011)	25ft Lineset

Standard Parts	
8-733-969-065	Remote Control

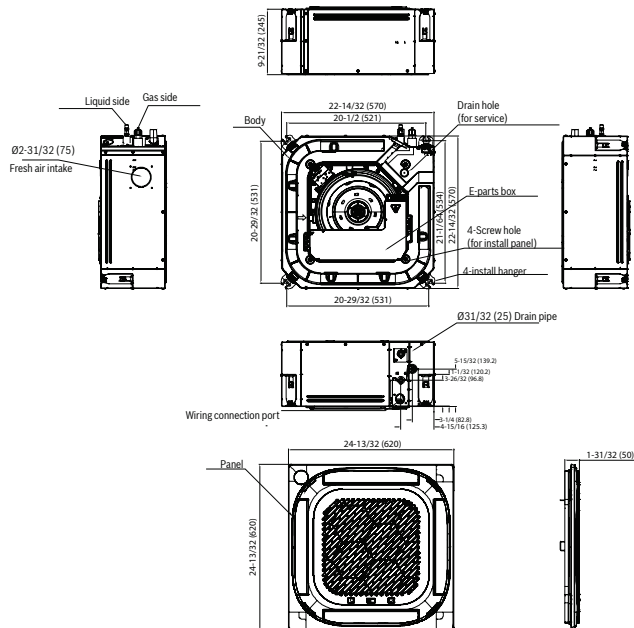
Climate 5000 Series - Gen 4

12K Btu Single Zone Regular System - 4-Way Cassette IDU



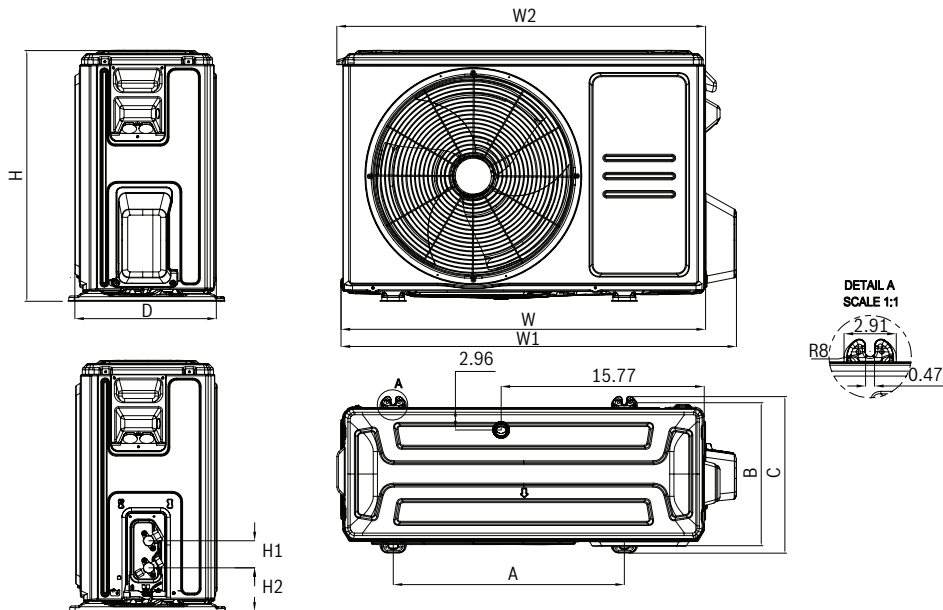
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Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU012-1AHCXD	22.44	22.44	9.65

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS012-1CSXRD	30.12	11.9	21.85	32.9	30.87	17.8	11.26	12.36	2.36	3.67

Climate 5000 Series - Gen 4

12K Btu Single Zone Regular System - 4-Way Cassette IDU


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Cooling Performance - Indoor Dry Bulb Temperature at 418 CFM																						
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)					
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	
-22°F (-30.0°C)	TC S/T PI	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	
-13°F (-25.0°C)	TC	10311	10325	10336	10336	10350	10932	10945	10956	10958	10971	11585	11600	11613	11615	11628	12283	12297	12310	12312	12327	
	S/T	0.48	0.56	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.77	0.54	0.63	0.72	0.72	0.82	0.57	0.67	0.76	0.76	0.87	
	PI	0.55	0.55	0.55	0.55	0.55	0.59	0.59	0.59	0.59	0.59	0.62	0.62	0.62	0.62	0.62	0.66	0.66	0.66	0.66	0.66	
-4°F (-20.0°C)	TC	11104	11119	11131	11131	11146	11773	11787	11799	11801	11815	12477	12493	12507	12509	12523	13228	13243	13257	13259	13276	
	S/T	0.48	0.57	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.78	0.54	0.64	0.72	0.72	0.82	0.57	0.67	0.77	0.77	0.87	
	PI	0.59	0.60	0.60	0.60	0.60	0.63	0.63	0.63	0.63	0.63	0.67	0.67	0.67	0.67	0.67	0.71	0.71	0.71	0.71	0.71	
0°F (-17.8°C)	TC	11456	11472	11484	11485	11500	12147	12161	12173	12176	12190	12873	12889	12904	12906	12921	13648	13663	13678	13680	13697	
	S/T	0.48	0.57	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.78	0.54	0.64	0.72	0.72	0.83	0.58	0.68	0.77	0.77	0.88	
	PI	0.61	0.61	0.62	0.62	0.62	0.65	0.65	0.65	0.65	0.65	0.69	0.69	0.69	0.69	0.69	0.73	0.73	0.73	0.73	0.73	
5°F (-15.0°C)	TC	11897	11913	11926	11926	11943	12614	12629	12642	12644	12659	13368	13385	13400	13402	13417	14173	14189	14204	14206	14224	
	S/T	0.49	0.57	0.65	0.65	0.74	0.51	0.60	0.69	0.69	0.78	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88	
	PI	0.64	0.64	0.64	0.64	0.64	0.68	0.68	0.68	0.68	0.68	0.72	0.72	0.72	0.72	0.72	0.76	0.76	0.76	0.76	0.76	
14°F (-10.0°C)	TC	11642	11658	11670	11671	11686	12344	12358	12370	12373	12387	13081	13098	13113	13115	13130	13869	13884	13899	13902	13919	
	S/T	0.49	0.57	0.65	0.65	0.74	0.52	0.61	0.69	0.69	0.78	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88	
	PI	0.60	0.60	0.60	0.60	0.60	0.64	0.64	0.64	0.64	0.64	0.67	0.67	0.68	0.68	0.68	0.71	0.72	0.72	0.72	0.72	
17°F (-8.3°C)	TC	11557	11572	11585	11585	11601	12253	12268	12280	12282	12297	12985	13002	13017	13019	13034	13768	13783	13798	13800	13817	
	S/T	0.49	0.57	0.65	0.65	0.74	0.52	0.61	0.69	0.69	0.79	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.78	0.78	0.88	
	PI	0.59	0.59	0.59	0.59	0.59	0.62	0.62	0.62	0.62	0.62	0.66	0.66	0.66	0.66	0.66	0.70	0.70	0.70	0.70	0.70	
23°F (-5.0°C)	TC	11689	11705	11717	11718	11734	12394	12408	12421	12423	12438	13134	13151	13166	13168	13183	13925	13941	13956	13958	13975	
	S/T	0.49	0.58	0.65	0.65	0.75	0.52	0.61	0.69	0.69	0.79	0.55	0.65	0.74	0.74	0.84	0.58	0.69	0.78	0.78	0.89	
	PI	0.60	0.61	0.61	0.61	0.61	0.64	0.64	0.64	0.64	0.64	0.68	0.68	0.68	0.68	0.68	0.72	0.72	0.72	0.72	0.72	
35°F (1.7°C)	TC	11954	11970	11983	11983	11999	12674	12689	12702	12704	12719	13431	13449	13464	13466	13481	14241	14256	14272	14274	14292	
	S/T	0.49	0.58	0.66	0.66	0.75	0.52	0.61	0.70	0.70	0.79	0.55	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89	
	PI	0.64	0.64	0.64	0.64	0.64	0.68	0.68	0.68	0.68	0.68	0.72	0.72	0.72	0.72	0.72	0.76	0.76	0.76	0.76	0.76	
41°F (5.0°C)	TC	11620	11636	11649	11649	11665	12321	12335	12348	12350	12365	13057	13074	13088	13091	13105	13844	13859	13874	13876	13893	
	S/T	0.49	0.58	0.66	0.66	0.75	0.52	0.61	0.70	0.70	0.80	0.56	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89	
	PI	0.52	0.52	0.52	0.52	0.52	0.55	0.55	0.55	0.55	0.55	0.58	0.58	0.58	0.58	0.58	0.62	0.62	0.62	0.62	0.62	
47°F (8.3°C)	TC	11287	11302	11315	11315	11330	11967	11982	11994	11996	12010	12682	12699	12713	12715	12730	13447	13461	13476	13478	13495	
	S/T	0.50	0.58	0.66	0.66	0.75	0.53	0.62	0.70	0.70	0.80	0.56	0.65	0.74	0.74	0.85	0.59	0.69	0.79	0.79	0.90	
	PI	0.40	0.40	0.40	0.40	0.40	0.42	0.42	0.42	0.42	0.42	0.45	0.45	0.45	0.45	0.45	0.47	0.47	0.47	0.47	0.48	
50°F (10.0°C)	TC	11206	11221	11233	11233	11248	11881	11895	11907	11909	11923	12591	12607	12621	12624	12638	13349	13364	13378	13381	13397	
	S/T	0.50	0.58	0.66	0.66	0.76	0.53	0.62	0.70	0.70	0.80	0.56	0.66	0.75	0.75	0.85	0.59	0.70	0.79	0.79	0.90	
	PI	0.41	0.41	0.41	0.41	0.41	0.43	0.43	0.43	0.43	0.43	0.46	0.46	0.46	0.46	0.46	0.49	0.49	0.49	0.49	0.49	
59°F (15.0°C)	TC	10961	10976	10987	10988	11003	11621	11635	11647	11649	11663	12316	12332	12345	12348	12361	13058	13072	13086	13088	13104	
	S/T	0.50	0.59	0.67	0.67	0.76	0.53	0.62	0.71	0.71	0.81	0.56	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91	
	PI	0.44	0.44	0.44	0.44	0.44	0.47	0.47	0.47	0.47	0.47	0.49	0.50	0.50	0.50	0.50	0.52	0.52	0.53	0.53	0.53	
68°F (20.0°C)	TC	10716	10730	10742	10742	10757	11362	11375	11386	11389	11402	12040	12056	12070	12072	12085	12766	12780	12794	12796	12812	
	S/T	0.51	0.60	0.69	0.69	0.78	0.54	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88	0.61	0.72	0.82	0.82	0.93	
	PI	0.47	0.47	0.47	0.47	0.47	0.50	0.50	0.50	0.50	0.50	0.53	0.53	0.53	0.53	0.53	0.56	0.56	0.56	0.56	0.56	
77°F (25.0°C)	TC	10471	10485	10496	10497	10511	11102	11115	11126	11128	11142	11765	11781	11794	11796	11809	12474	12488	12501	12503	12519	
	S/T	0.52	0.61	0.69	0.69	0.79	0.55	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89	0.62	0.73	0.83	0.83	0.94	
	PI	0.50	0.50	0.51	0.51	0.51	0.53	0.54	0.54	0.54	0.54	0.57	0.57	0.57	0.57	0.57	0.60	0.60	0.60	0.60	0.60	
82°F (27.8°C)	TC	10664	10679	10690	10691	10705	11307	11321	11332	11334	11347	11983	11998	12012	12014	12027	12705	12719	12732	12734	12750	
	S/T	0.53	0.62	0.70	0.70	0.80	0.56	0.66	0.75	0.75	0.85	0.59	0.70	0.79	0.79	0.90	0.63	0.74	0.84	0.84	0.95	
	PI	0.58	0.58	0.58	0.58	0.58	0.62	0.62	0.62	0.62	0.62	0.65	0.65	0.66	0.66	0.66	0.69	0.69	0.69	0.69	0.70	
86°F (30.0°C)	TC	11235	11250	11262	11263	11278	11912	11926	11938	11940	11954	12624	12640	12654	12657	12671	13384	13399	13413	13416	13432	
	S/T	0.53	0.63	0.71	0.71	0.81	0.57	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91	0.64	0.75	0.85	0.85	0.97	
	PI	0.72	0.72	0.72	0.72	0.72	0.76	0.76	0.76	0.76	0.76	0.77	0.81	0.81	0.81	0.81	0.86	0.86	0.86	0.86	0.86	
95°F (35.0°C)	TC	12519	12535	12549	12550	12566	13273	13289	13302	13305	13320	14066	14084	14100	14103	14118	14914	14930	14946	14948	14967	
	S/T	0.53	0.63	0.71	0.71	0.81	0.57	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91	0.64	0.75	0.85	0.85	0.97	
	PI	1.03	1.03	1.03	1.03	1.03	1.09	1.09	1.09	1.09	1.09	1.10	1.16	1.16	1.16	1.16	1.23					

Climate 5000 Series - Gen 4

12K Btu Single Zone Regular System - 4-Way Cassette IDU



BOSCH

Heating Performance - Indoor Dry Bulb Temperature at 418 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	/	/	/	/	/	/	/	/	/	/
-13°F(-25.0°C)	5435	5327	5300	5279	5231	0.92	0.93	0.93	0.93	0.94
-4°F(-20.0°C)	6871	6734	6700	6673	6613	0.98	0.99	0.99	0.99	1.00
0°F(-17.8°C)	7646	7493	7456	7426	7359	1.00	1.00	1.01	1.01	1.01
5°F(-15.0°C)	8615	8442	8400	8366	8291	1.02	1.02	1.03	1.03	1.03
14°F(-10.0°C)	9307	9121	9075	9039	8957	1.05	1.06	1.06	1.06	1.07
17°F(-8.3°C)	9537	9347	9300	9263	9179	1.06	1.07	1.07	1.08	1.08
22°F(-5.6°C)	9954	9755	9706	9667	9580	1.07	1.07	1.08	1.08	1.08
27°F(-2.8°C)	10370	10163	10112	10071	9981	1.07	1.08	1.08	1.08	1.09
32°F(-0.0°C)	10786	10570	10518	10475	10381	1.07	1.08	1.09	1.09	1.09
35°F(1.7°C)	11036	10815	10761	10718	10622	1.08	1.09	1.09	1.09	1.09
37°F(2.8°C)	11760	11525	11468	11422	11319	1.10	1.10	1.11	1.11	1.11
42°F(5.6°C)	13572	13300	13234	13181	13062	1.14	1.15	1.15	1.16	1.16
44.6°F(7.0°C)	14296	14010	13940	13884	13760	1.16	1.17	1.17	1.17	1.18
47°F(8.3°C)	15383	15075	15000	14940	14806	1.19	1.20	1.20	1.20	1.21
52°F(11.1°C)	14899	14601	14528	14470	14340	1.10	1.11	1.11	1.12	1.12
57°F(13.9°C)	14415	14127	14056	14000	13874	1.02	1.02	1.03	1.03	1.03
59°F(15.0°C)	14221	13937	13867	13812	13688	0.98	0.99	0.99	0.99	1.00
62°F(16.7°C)	13931	13652	13584	13530	13408	0.93	0.94	0.94	0.94	0.95
64.4°F(18.0°C)	13699	13425	13358	13304	13185	0.89	0.90	0.90	0.90	0.90
69.8°F(21.0°C)	13176	12913	12848	12797	12681	0.80	0.80	0.81	0.81	0.81
75.2°F(24.0°C)	12653	12400	12338	12289	12178	0.71	0.71	0.71	0.71	0.72

Bolded values were tested. All other values are simulated and for reference only.